

# Bridging the Past and Present: Sustainable Water Management Technologies Through the Lens of Hydro-Science in Ancient India

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## Abstract

This article delves into the ancient Indian techniques of sustainable water resource management and the corresponding modern-day approaches of hydro-science towards sustainability. This article takes a comprehensive look at the traditional methods of water management that have been practised in India for centuries and highlights their relevance in the present-day context. The ancient techniques of water management in India, including rainwater harvesting, groundwater recharge and canal systems, were developed based on a deep understanding of the local ecosystems and the water cycle. These practices were designed to promote the sustainable use of water resources and maintain a delicate balance between water availability and demand. In recent times, with the increasing global awareness of sustainability, there has been a renewed interest in these traditional practices which have been developed with a focus on reducing environmental impacts, conserving natural resources and promoting social and ecological equity which includes green technologies, circular economies and sustainable development goals and are rooted in the principles of sustainability that were ingrained in ancient India. In summary, this research contributes to emphasize the need to preserve traditions and integrate them with modern technology to achieve sustainable growth and ensure a sustainable future for all.

## Keywords

Sustainable water resource management, technology and sustainability, ancient Indian techniques, integration of tradition, hydro-science

## Introduction

According to the Vedas, life on earth emerged from ‘*Apah*’, which means ‘Water’. The Rig Veda mentions several gods associated with water, including Apas, Varun, Indra and Pranjay. The divine serpent on

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which Narayana rests, Adi Shesha, is also seen as a representation of water. The hymns of Rig-Veda, X.B2.1 and X.121.1 describe the beginning of the creation which is marked by the emergence of water and the cosmic golden egg, also known as '*Hiranyagarbh*' - '*hiranyagarbhaḥ samavartatāgre bhūtasya jātah patir ekāśīt sa dādāhāra prthivīm ghām utemām kasmai devāya haviṣā vidhema*' (R.V.X,121.1) (Sarasvati, 2009).

The importance of water and the perspective of hydro-science, hydro-mechanics and water management in ancient India is of utmost importance given the challenges of climate change, environmental destruction, water scarcity and extreme hydrological events. With a population of over a billion people, the abuse and overuse of natural resources in India are leading the country towards political, social and environmental catastrophes. Experts, specialists and policymakers in the field of soil and water conservation in developing countries have realized that their past strategies have failed and it is high time to accept the pressing need to explore the knowledge about water management in ancient Indian texts. As a result, there is increasing recognition of the value of traditional knowledge regarding water and the various natural resources which were previously ignored. Traditional methods and knowledge can provide a suitable starting point for developing alternative solutions that are both adaptable and environmentally sustainable.

The Harappan (or Indus Valley) (3000–1500 BCE) civilization was among the earliest most technologically advanced civilizations of ancient times, withholding largest geographical area and exemplified level of scientific and societal development in the protohistoric Indian subcontinent. According to Jansen (1989), the Harappans were known for their reverence for water, worshipping the rivers daily by giving them divine status. The urban centres of the civilization were designed with advanced civil and architectural engineering, including sophisticated drainage and wastewater management systems. It is noteworthy that not only pure water but also wastewater management technologies have adapted to the socioeconomic, sociocultural and religious practices of individual societies throughout the ages of civilizations (Feo & Napoli 2007; Lofrano & Brown, 2010; Sorcinelli 1998; Wolfe, 1999).

The ancient Indian scriptures of the Vedic time contain extensive knowledge about hydro-science. References to hydrology can be found in the Rig-Veda, Yajur-Veda, Atharv-Veda, Puranas like Ramayana, Mahabharata and other texts such as Brihat-Samhita, Arthashastra and Meghamala. The texts provide invaluable verses that cover topics such as the cycle of water, formation of clouds, flow of streams, level of underground water, hydraulic machinery, different hydro-structures and eco-friendly approaches for managing wastewater. But, until recent times, this vast amount of knowledge remained largely unexplored. M. Visvesvarayya, the architect of water management in modern India, drew heavily from the wisdom of ancient civilizations, particularly Indian, in his pioneering works.

During the Mauryan era, the rainfall patterns across different regions of India were studied and identified and additionally, the Mauryans are credited with the creation of the world's first observatory (Srinivasan, 1976). The Mauryan Empire, which existed from 322-185 BCE, is considered to be the first 'Hydraulic Civilization' due to their innovative use of dams with the help of spillways which was generally addressed as *Ahars and Pynes* (Singh et al., 2020). Building upon the ancient knowledge of traditional water management systems and the use of natural materials, modern India is committed to developing a more sustainable and resilient water management system by leveraging modern technology in rainwater harvesting. The construction of dams in India during ancient times stands as a testament to the advanced water management techniques of the civilization. Presently, India boasts one of the world's largest numbers of dams, with a total storage capacity exceeding 300 billion cubic meters. The Integrated Waters Management Program is one of the initiatives implemented by the government, which is a revised version of the Drought Prone Areas Programme (DPAP), Desert Development Programme (DDP) and

Integrated Wastelands Development Programme (IWDP) under the Department of Land Resources and aims to achieve integrated planning, sustainable outcomes, and optimum utilization of resources.

## Historical Overview of Hydro-Science in Ancient India

Besides the great tradition of spirituality and holism, the Ancient Indian Civilization was enriched by the extensive growth of science and technology including the ‘Hydro-Science’ which was not only advanced in terms of technology usage but also credible in respect of environmental sustainability. It dates back to the Harappan civilization (approximately 3000–1500 BCE) to the Vedic Period (approximately 1500–500 BCE) and to the Maurayan empire (322–185 BCE). Ancient India possessed expertise in hydrological, hydraulic and related forms of engineering, which encompassed a deep understanding of the water cycle and its interconnected processes. Recent explorations and findings have revealed this knowledge, particularly through the Vedas, including the Rig-Veda, Yajur-Veda and Atharv-Veda.

As this research delves further into the testimonials of the hydrological works founded in ancient Indian literature, which includes mythology, it has discovered a multitude of intriguing facets of India’s scientific achievements. These contributions span across the entire geographical region of the Indian subcontinent, from the Indus River in the east to the Cauvery River in the south.

## *Sustainability and Resilience of Ancient Indian Water Management Practices*

The formation of social and cultural societies, as well as the establishment of agriculture and permanent settlements, created a distinct relationship between humans and water. Scholars such as Scarborough (2003) and Ortloff (2009) have examined that how the water management techniques imparted their regime in ancient social structures and used examples from both the eastern and western hemispheres to support their arguments. In a comprehensive review of the history of wastewater management, Lofrano and Brown (2010) have discussed the evolution of sanitation across various civilizations, including the ancient Indus Valley civilization.

India’s civilization, like many others, grew alongside rivers and deltas, which continue to symbolize the traditional culture (Nair, 2004). The Harappan civilization, which thrived about 5000 BCE (Dixit et al., 2018) or from 2600–1900 BCE (Chase et al., 2014), is famous for its meticulously designed urban areas, which included public and private bathing facilities, an intricate network of underground drains which is built with perfectly arranged bricks with an elaborate water management system consisting of number of wells and reservoirs (Sharma & Shruthi, 2017). The Harappan people achieved a remarkable fusion of civil, material and architectural sciences, creating one of the most intelligent urban centres of its time. The advanced irrigation systems in ancient India played a significant role in the growth of deeply complex urban culture and societies. The Harappan Civilization largely named as Indus Valley civilization demonstrated a high level of expertise in hydraulic engineering and is recognized as the pioneer of the world’s earliest flush toilets. Kenoyer (2003) contends that ‘no other city in the ancient world had developed such a sophisticated water and waste management system. Even during the Roman Empire, some 2000 years later, these kinds of facilities were limited to upper-class neighbourhoods’.

Dholavira, one of the prominent city in the Indus Valley civilization, had sophisticated and well-managed water management systems that included a complex network of step wells, channels and reservoirs. The city was encircled by 16 large reservoirs, some interconnected, that measured 7m deep and 79m long, accounting for approximately 10% of the city’s area (Iyer, 2019). The people of Dholavira

exhibited exceptional engineering skills by successfully conserving every drop of water in their arid environment (see Fig. 1). A rectangular step well, which is 73.4m long, 29.3 meters wide and 10m deep, was recently discovered in Dholavira which is three times larger than the famous Great Bath in Mohenjo-Daro. The hydraulic engineering capabilities of the Harappans of Dholavira are evident in the systems they developed for water conservation, harvesting and storage, especially considering the technological limitations of their time (Baba, 2018).

‘Lothal’ which means the ‘mound of the dead’ is another ancient city of the Harappan civilization situated at the doab of the Bhogavo and Sabarmati rivers (see Fig. 2). It is also known as the ‘harbour’



**Figure 1.** Water Reservoir at Dholavira.

**Source:** India.com (2015).



**Figure 2.** Lothal.

**Source:** Wikipedia.org (2010).

city due to its significance as a trading centre for civilization. Archaeologists believe that the trapezoidal structure at Lothal, which measures 209.30m on the eastern embankment, 212.40m on the western one, 36.70m on the northern one and 34.70m on the southern one, is the world's first known dockyard. This structure is evidence of advanced maritime activities during ancient times (Nigam et al., 2016).

The Harappans were involved in extensive agriculture and used a sophisticated irrigation system that included storage systems, wells and water-harvesting techniques at low cost like small bunds and check dams. Evidence shows that the Harappans constructed such structures using rock-cut pieces and boulders to divert and reduce water velocity, thus reducing siltation in main reservoirs, as seen in Dholavira. The *Ahar-Pyne* system, a participatory irrigation management with sophisticated rainwater harvesting structure used during the Mauryan era, was actually a sustainable and cost-effective rainwater harvesting technique. To manage their water resources, the Harappans employed various techniques such as contouring, bunding, terracing, benching, *Gabarbands* or the dams built of stone for storing and controlling water, and canals. These techniques were crucial to ensuring the sustainability of their agricultural practices in the arid regions of the Indus Valley (McKean, 1985). In addition to the techniques mentioned earlier, the Indus Valley civilization practised Rabi irrigation, which mainly relied on wells, canals and floodwater canalization through ditches for irrigating the dry Rabi crops. This technique was common and was used alongside other practices like small-scale irrigation, lift irrigation and pond irrigation when flood waters were not available. In the Mauryan era, the *Ahar-Pyne* system was also utilized as a rainwater harvesting technique to supplement the water supply during dry periods.

In the Vedic age, people recognized the significance of storing water from hilly regions and transporting it to far-off areas through canals. This practice allowed them to utilize water resources more effectively and sustainably (Bhattacharya and India, 2012). The Rigveda includes multiple verses that highlight the significance of using water from ponds and wells for agricultural purposes such as verses I, 23.18 and V, 32.2. Additionally, verse VIII, 3.10, demonstrates the construction of various artificial canals by engineers called *Ribhus* for the irrigation of desert areas. The Rigveda also emphasizes the importance of utilizing water efficiently since it is a natural gift that benefits all. This highlights the understanding of the importance of conserving water resources and using them wisely for the greater good as seen in verses VIII, 49.6 and X, 64.9. The Rigveda also mentions irrigation methods such as wells through verse X, 25; canal digging in verse X, 75 and canals through using the word '*kulya*' in verse X, 99. The word '*kulya*' is also used in the Mahabhasya of Patanjali (150 BCE).

The Rigveda describes an ancient device known as '*asma-cakra*,' which was a wheel made of stones that were used to raise water. The device operated with the assistance of a leather strap, is an example of the innovative methods used by ancient Indian civilizations for water management. It also mentions the '*Ghatayantra*' or '*Udghatana*,' which is a drum-shaped wheel with a pair of endless ropes around it, with earthen pots fixed at equal intervals. This waterlifting wheel is also known as '*Noria*' in Arabic literature. The '*asma-cakra*' refers to a stone pulley, which was used to pull the buckets filled with water around and emptied into channels using strings made of leather. The Arthashastra also makes reference to a method of watering crops, where a mechanical apparatus named '*Udghatam*' is utilized to lift water from natural sources like lakes, wells, tanks or rivers for irrigation purposes (Srinivasan, 1970).

Yajurveda verses VI, 100.2 and VII, 11.1 describe how individuals can bring water to desert areas using wells, ponds, canals etc., and how people should prepare for natural calamities like droughts and floods. According to Verse XX, 77.8 in the Atharvaveda, it is advised that the king should build canals over mountains to ensure the availability of water for agriculture and various other needs of his people. The Atharvaveda demonstrates about managing drought by making effective use of existing water resources which highlights the significance of using water efficiently to lessen the severity of drought. Verse 2.3.1 of the Atharvaveda provides instructions for the proper management of various water bodies



such as brooks, wells, and pools to reduce drought intensity and water scarcity. According to Kenoyer (2003), there is evidence to suggest that both Harappa and Mohenjo-Daro were inhabited during the Vedic Period, indicating the necessity for additional research and archaeological inquiries into this specific era (1500-500 BCE).

According to Shaw and Sutcliffe (2001), the Sudarsana Lake in Junagadh was originally built by Pusyagupta, who served as a provincial governor under Chandragupta Maurya. Later on, Ashoka made improvements to the lake by adding more conduits to enhance its functionality. Sutcliffe and Shaw also discovered several dams with spillways in Sanchi, Madhya Pradesh, which were constructed based on principles of water balance (Kumar & Katara, 2020). In addition, the Western Ghats' Naneghat has an ancient water harvesting system that includes drinking water tanks, and the Porumamilla tank inscription of Bhaskara Bhavadura (1369 A.D.) in South India details the construction of dams and tanks. These findings suggest that the hydro-science not only existed but also flourished in ancient times.

Shaw and Sutcliffe suggest that the engineers who constructed the Sanchi dam system were likely experienced in reservoir irrigation and had an advanced understanding of hydraulic technology. Based on the strong correlation between the volume of runoff and reservoir in the Sanchi region, it can be inferred that ancient Indians possessed a deep understanding of water balance, which was likely gained through extensive observation and knowledge of local conditions.

Despite their findings, Shaw and Sutcliffe note that a deep thorough understanding of the system of ancient Indian irrigation practices could have been achieved if the Sanchi reservoir complex had received more attention during the Vedisa excavations. Bhandarkar's 1914 excavation of the 'Heliodorus' pillar area in Vedisa (currently known as Vidisha, Madhya Pradesh) revealed the canal-remains from 300 BCE that likely drew water from the Betwa river.

Singh et al. (2020) cite the *Ahar-Pyne* system of the Mauryan Empire as a remarkable ancient hydraulic structure utilized for managing rainwater harvesting and irrigation. Even in the recent days, this system is extensively practised in certain areas of Chhota Nagpur and southern Bihar. The Pynes channels are constructed to harness river water

flow from hilly areas, while Ahars serve as catchment areas with three-side embankments to collect rainwater from Pynes. Numerous Ahars are fed by Pynes, and multiple distributaries are created from both systems for irrigation purposes. The *Ahar-Pyne* system is particularly well-suited for areas that experience limited rainfall, have rugged and rocky terrain, loose sandy soils or heavy clay and steep slopes. The system also acted as a process for mitigating floods. Pynes come in different sizes, with some originating from Ahars and used for irrigating cultivable fields, while others originate from rivers and are known as '*dasianpynes*,' irrigating many thousand acres of land. Under the 'Jal Jeevan Hariyali' program, the Bihar government has initiated the refurbishment of traditional water bodies, highlighting the significance of the *Ahar-Pyne* (see Fig. 3) system in contemporary times for water harvesting.

According to the Arthashastra, it was the responsibility of the king to ensure the construction of reservoirs (*sétu*) that could be filled with two sources—perennial water or water drawn from other sources. Alternatively, the king could provide essential resources like land, roads, timber and other materials to the subjects to build reservoirs independently.

In the Sangam Period (300 BCE to 300 AD) in southern India, people constructed tanks (*ery* in Tamil) and lotus ponds (*tamaraiikulam* in Tamil) as rainwater harvesting structures for the irrigation of paddy fields and fishing respectively. In the first century AD, the Chola King Karikalan initiated the construction of Grand Anicut (Kallanai Dam) on the Cauvery River to safeguard downstream populations from floods and provide irrigation water to the Cauvery delta region. The Grand Anicut, which is still functional, is considered the world's oldest dam in continuous use, as well as the first one built in India and the fourth oldest dam worldwide (see Fig. 4).



**Figure 3.** Usage of *Ahar-Pyne* System in Jal Jeevan Hariyali Mission, Bihar, India.

**Source:** Yashwant (2019).



**Figure 4.** The Grand Anicut (Kallanai Dam).

**Source:** Kumar (2014).

Around 550 AD, the *Brihat Samhita* text documented references to the arrangement of ponds and protection of bank measures, including plantation with pitching and sluicing. The text indicates that a pond aligned from east to west is more effective in retaining water for a longer time, while one aligned from north to south tends to lose water due to waves created by the wind. Additionally, the text advises constructing a spillway on one side of the pond with the passage paved with stones to serve as an outlet for water. These references in the *Brihat Samhita* demonstrate a high level of knowledge and understanding of hydraulic technology during the time.

## Early Forms of Wastewater Treatment in India

Over time, the management of sanitation and wastewater has been a significant socio-environmental concern, and ancient Indian civilizations were able to develop advanced technological layouts by combining their understanding of hydraulic systems of management with innovations in materials and structures. The cities of Harappa were among the earliest and most developed urban centres globally, showcasing impressive architectural and civil expertise. During 2500 BCE, Harappa and Mohenjo-Daro were equipped with the world's first urban sanitation systems (Webster, 1962). According to a thorough review of the history of wastewater management by Lofrano and Brown (2010), the Indus Valley civilization was the first ancient society to have implemented adequate systems for treating wastewater. The Harappan civilization's first urban sites were characterized by their wastewater management and sanitation (Kenoyer, 1991). The drainage and sewage systems were intricate networks in Harappa and Mohenjo-Daro and the primary elements of wastewater disposal included latrines, soak pits, cesspools, pipes and channels (Fardin et al., 2013).

The drainage channels in the ancient Harappan cities were covered with cut stones and bricks, and all houses were connected to them. The wastewater from the household was collected using terracotta pipes with a tapered shape and then routed to small sumps where sedimentation occurred, and larger contaminants could be removed. That was likely the first recorded attempt at wastewater treatment. The U-shaped pipes used for collecting wastewater in ancient India were made of well-burned bricks and were set in clay mortar. They were covered with materials like flagstones, wooden boards or brick slabs, and could be easily removed for the purpose of cleaning. These pipes were an important precursor to modern vitrified clay spigot-and-socket sewer pipes (Gray, 1940). To ensure the easy maintenance and cleaning of drainage systems, the channels were designed in a way that allowed for the removal of cut stones and bricks. Cesspits were also installed at the intersection of multiple drains to prevent blockage of the systems. (Wright, 2010).

The ancient cities of Indus Valley Civilization were equipped with advanced sewage systems that included multiple flushing lavatories (Pruthi, 2004). The hydraulic engineering in the Harappan civilization is exemplified by remarkable structures such as the Great Bath in Mohenjo-Daro (see Fig. 5), the 16-reservoir system of Dholavira and the dockyard.

At Kalibangan, U-shaped channels which were made of bricks of terracotta or wood were used to connect toilets and bathroom outflows to decentralized sewage systems. These effluents were then directed to a jar placed in the main street (Chakrabarti, 1995). In Banawali, a similar system was used to collect wastewater, and the effluents were directed into drains constructed using clay bricks before being discharged into the jars. (Bisht et al., 1984).

The Atharva Veda, Charaka Samhita, Susruta Samhita and Ashtanga Hridaya Samhita (ninth century AD), as well as Bhava Prakash (sixteenth century AD), are ancient standard texts that discuss water quality, which is an important aspect of Ayurveda—the science of life. The Rig Veda states: '*Pavātāvanti patayanti vidyut udoṣadhīr jihvā te pinvate svāhā Irā viśvam bhuvanāya jāyate yat parjanyaḥ prthivī retasāvati*' (R.V., V, 83.4), which means that tree plantation, forest conservation and Yajnas (religious rituals) are ways to purify the environment and improve the quality of water for the well-being of mankind (Jain et al., 2018).

The Yajurveda verse (I,12) discusses water contamination and suggests that *Yajna* or heat, and sun rays can purify water by breaking down substances into smaller particles. The Mahabharata (XII,184.31 and 224.42) describes the various qualities of water based on taste. The Brihat Samhita verse (54.104) notes that soil colour can indicate water quality, stating that brown soil can make water alkaline, pale white soil can create saltwater and blue soil can make water pure and sweet. Brihat Samhita suggests





**Figure 5.** Great Bath of Mohenjo-Daro.

**Source:** Ghani (2019).

that: ‘*Ajjana Musto śīraiḥ śarājakośā takāmalakacūṇaiṃḥ kataka phala samāyuktair yogah. kūpe pradātavyah*’ (Vr.S.54.121), which means that a mixture of specific ingredients, such as *Anjanam* (kajal), *Nagarmodha*, *Khas*, Powder of *Torayi* and *Amala*, along with clear nuts, can be added into a well to improve the quality of water. This mixture can help clear the water, improve its taste and make it aromatic with other beneficial qualities. The verse ‘*Kalusham katu kam lavanam virasam salilam yadiva ashubhagandhi bhavet tad anena bhavati amalam surasam susugandhi ganairaparescha yutam*’ (Vr.S.) suggests that this method can be useful for the water that is turbid, has a bad taste and unpleasant odor.

The Bhava Prakash text with its 86 verses named *Vari Vargah* provides insights into different aspects of water, including its types and properties. For example, verses 26–31 describe three types of water: *Jangala* (found in areas with sparse trees), *Anupam* (found in areas with abundant trees) and *Sadharanam* (found in areas with moderate trees). *Jangala* water is described as saltish and helpful in eliminating *Pitta* and *Kapha*, *Anupam* water is tasty and viscous, promoting *Kapha*, while *Sadharana Jalam* is sweet and eliminates *Tridosha* (the three diseases) (Jha, 1988).

## The Evolution of Technologies from Ancient to Modern for Sustainable Water Management

The issue of sustainability has gained significant importance in contemporary India, and it is noteworthy as discussed above that several techniques and principles for sustainable living have their origins in ancient Indian culture. Throughout its history, Indian society has placed emphasis on utilizing natural resources responsibly and conserving the environment, understanding that the maintenance of a healthy ecosystem is crucial for the welfare of both individuals and society at large.

As India grapples with the challenges of climate change, pollution and water scarcity, these ancient techniques are being revisited and combined with modern technologies to create innovative solutions for

sustainable development. Sustainable water management is a particularly critical area, as water is a precious resource that is under threat from overuse, pollution and climate change.

By drawing on ancient Indian techniques such as rainwater harvesting, traditional water management systems and the use of natural materials, modern India is working towards a more sustainable and resilient water management system. Additionally, modern technologies such as smart water meters, remote sensing and GIS are being employed to optimize water distribution, reduce waste and detect leaks. The integration of ancient techniques and modern technology is helping India to address the challenges of sustainability in a comprehensive and effective way. By focusing on sustainable water management, India can ensure that its citizens have access to clean, safe, and reliable water while preserving this vital resource for future generations.

### *Rainwater Harvesting*

Rainwater harvesting is an ancient Indian technique that has been used for thousands of years to collect and store rainwater. In ancient India, this technique was particularly important in regions with arid climates, where water was scarce and communities had to rely on rainfall for their water needs.

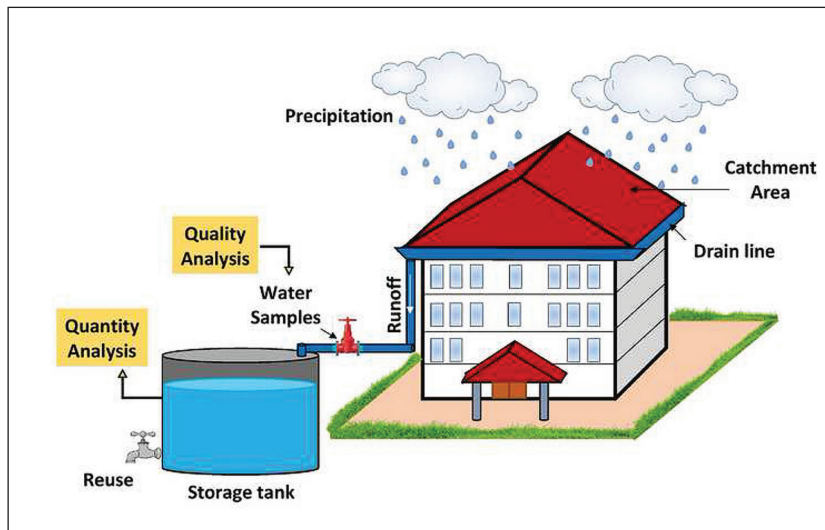
Rainwater harvesting is not only practised in arid areas but also in other parts of the country to harvest water and reduce the dependency on groundwater resources. During ancient times be it the Harappan Civilization or the Mauryan Empire, people frequently engaged in rainwater harvesting, which involved the utilization of tanks and reservoirs to accumulate and preserve water throughout the monsoon season. These tanks were often built underground or into hillsides and were designed to prevent evaporation and contamination. The water stored in these tanks was then used for irrigation, drinking and other household needs throughout the year. The work of Mane and Shinde (2014) describes two main ways of harvesting rainwater, that is, Runoff Harvesting and Rooftop Rainwater Harvesting (RTRWH).

*Runoff Harvesting:* This process of rainwater harvesting involves capturing rainwater that flows away as surface runoff in urban areas. This water can be used for recharging aquifers through appropriate methods and there are evidences of certain technological similarities with the *Ahar-Pyne* system of the Mauryan Empire and the Sanchi Dam System. However, urbanization increases surface runoff by creating more impervious surfaces like pavement and buildings that prevent water from percolating down through the soil to the aquifer. Instead, it is forced directly into streams or runoff drains, leading to problems like erosion and siltation. Increased runoff reduces groundwater recharge, lowers the water table and exacerbates droughts for farmers and others who rely on wells for water (see Fig. 6).

*Rooftop Rainwater Harvesting (RTRWH):* The second method is rooftop rainwater harvesting (RTRWH), which involves storing rainwater when it falls on rooftops. The roof of a house or building serves as the surface to collect rainwater, which can be directed to an artificial recharge system or stored in a tank (see Fig. 7). This low-cost approach is highly efficient in enhancing the groundwater level of regions that are dry, hilly, urban or coastal in nature and ensures the availability of water in the future.

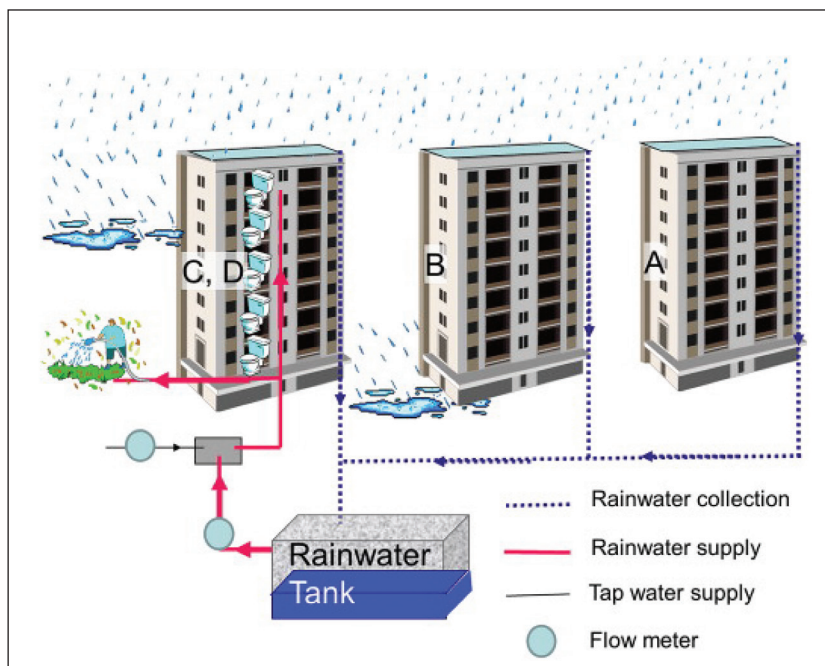
### *Hydropower Generation*

The construction of dams in India from ancient times was a testimony to the advanced water management techniques of the civilization. Presently, India possesses an extensive collection of dams, comprising a total storage capacity exceeding 300 billion cubic meters, which is among the highest in the world.



**Figure 6.** Runoff Rainwater Harvesting Flowchart.

**Source:** Anchan and Prasad (2021).



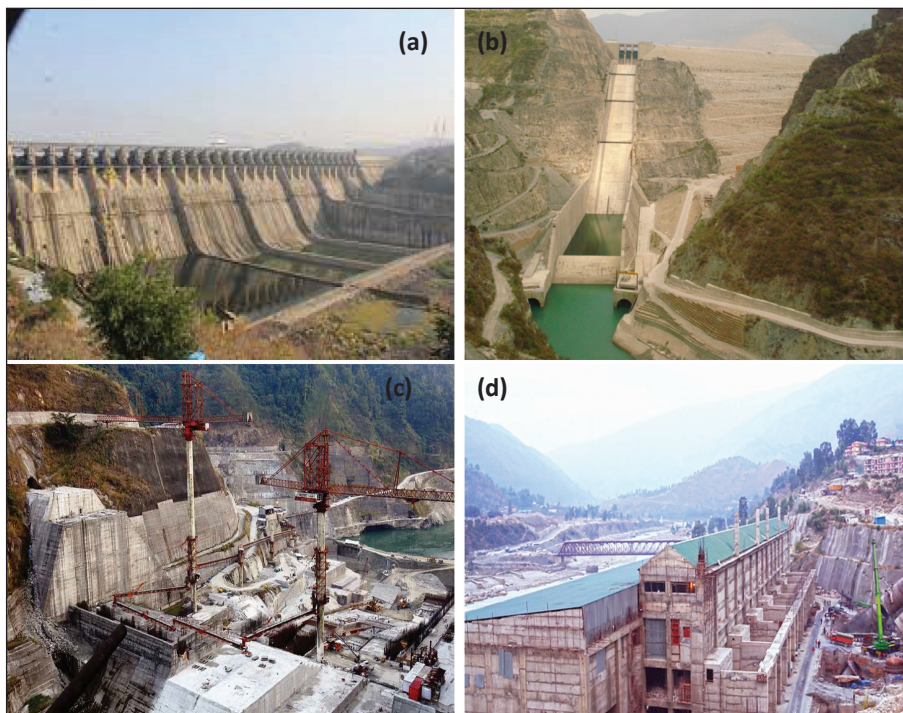
**Figure 7.** Rooftop Rainwater Harvesting (RTRWH) Flowchart.

**Source:** Mun and Han (2012).

Approximately 45,000 dams (Bid et al., 2018) with an elevation greater than 15m have been constructed across the globe, and a significant 73% of them were built in the last five decades. The first-ever known dam was built in 2900 BC to safeguard the city of Memphis from floods. In India, the first administrative records are found in Rudradamana's Girnar inscription, which mentions the construction of a dam at Sudarshana Lake in Saurashtra in 150 BCE by Chandragupta Maurya's governor, Pusyagupta. China's Yalong Jiang River boasts the tallest dam in the world at 305m, followed by India's Tehri Dam, which stands at 261m on the Bhagirathi River, Tehri, Uttarakhand. The longest dam in India is the Hirakud Dam, which spans about 26km across the Mahanadi River in Odisha state. Almost half of the world's rivers are obstructed by at least one large dam with an elevation greater than 15m. Developing countries and international organizations, such as the World Bank, have invested significantly in dam construction, believing that it can lead to development, reduce poverty and increase irrigation and hydroelectricity production.

Keeping in mind the ancient techniques of the construction of dams, modern Indian approaches to hydroelectricity generation have spiked in the last few decades.

India has been investing in modern hydropower projects to meet the increasing demand for electricity and reduce the country's dependency on fossil fuels. Here are some recent hydropower projects in India (see Fig. 8):



**Figure 8.** (a) Sardar Sarovar Dam; (b) Tehri Dam; (c) Subansiri Lower Hydroelectric Project; (d) Rampur Hydroelectric Project.

**Source:** (a) Wikipedia.org (2018); (b) Wikipedia.org (2008); (c) Probe (2022); (d) of Environmental Justice (2022).



- **Kishanganga Hydroelectric Project:** Located in Jammu and Kashmir, the Kishanganga Hydroelectric Project has an installed capacity of 330 MW. It was completed in 2018 at a cost of around ₹5,900 crores.
- **Sardar Sarovar Dam:** Located on the Narmada River in Gujarat, the Sardar Sarovar Dam is among the major hydroelectricity ventures in India, possessing an installed capacity of 1,450 MW. Despite numerous years of controversy and protests, the project was finished in 2017.
- **Tehri Dam:** The Tehri Dam, situated on the Bhagirathi River in Uttarakhand, has an installed capacity of 1,000 MW and is recognized as one of the world's highest dams, with a height of 260m.
- **Subansiri Lower Hydroelectric Project:** The Subansiri Lower Hydroelectric Project is being built on the Subansiri River in Arunachal Pradesh. With an installed capacity of 2,000 MW, it is expected to be one of the largest hydropower projects in India once completed.
- **Rampur Hydroelectric Project:** The Rampur Hydroelectric Project is located on the Satluj River in Himachal Pradesh. It has an installed capacity of 412 MW and was completed in 2014.

These modern hydropower projects in India have helped to increase the country's installed hydropower capacity and provide clean and renewable energy to meet the rising demand for electricity.

## Recent Government Schemes in Sustainable Water Management

The Integrated Water Management Program is a modified version of the earlier Drought Prone Areas Programme (DPAP), Desert Development Programme (DDP) and Integrated Wastelands Development Programme (IWDP) of the Department of Land Resources. The aim of this consolidation is to achieve optimum use of resources, sustainable outcomes and integrated planning. The program was launched in 2009–2010 and is being implemented in accordance with the Common Guidelines for Watershed Development Projects 2008. The IWMP's primary objectives are to restore the ecological balance by harnessing, conserving and developing degraded natural resources such as soil, vegetation and water. The program's outcomes include preventing soil erosion, regenerating natural vegetation, harvesting rainwater and recharging the groundwater table. These measures enable multi-cropping and the introduction of diverse agro-based activities that help provide sustainable livelihoods to people living in the watershed area.

## Conclusion

Water, one of the five gross elements (*'Panchbhut'*), was a source of inspiration for ancient humans, as it is not only precious but also a critical natural resource that is intimately linked to our survival and societal and cultural developments. Water holds significant cultural and religious value in Hinduism, where it is considered a sacred element and a source of life.

In addition to spiritual upbringing, ancient India also made magnificent advancements in the fields of science and technology, particularly in water conservation, harvesting and irrigation. Incorporating ancient water wisdom at all levels of hydro-science could significantly enhance existing knowledge of water conservation, harvesting and irrigation. The modern perspective on water often involves viewing it in a demand-supply paradigm as a commodity that can be traded for profit, with a focus on managing



its distribution and storage to meet demand. This narrow perspective has resulted in human rights over water being usurped by vested interests when demand exceeds supply.

Although the reason for the decline of the Harappan Civilization is still unclear, it is believed that a change in the course of the river and a reduction in the availability of water for the agricultural system ultimately caused people to migrate to the Gangetic plain, leaving Harappa deserted and abandoned. As one of the most significant occurrences, the flash flood in Kedarnath in 2013 serves as a poignant reminder of the disastrous consequences that can arise due to poor management of water resources. All of these observations emphasize the critical significance of water as a precious and priceless resource, not only in the past and present but also for the future, and it must be managed responsibly.

Therefore, the ancient knowledge of sustainability and wisdom about water management must be acknowledged and utilized, with a focus on our connection to nature rather than solely on capitalist demands. By integrating this wisdom into modern practices, a sustainable and alternative development approach can be established for the betterment of the future.

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### References

- Anchan, S. S., & Prasad, H. S. (2021). Feasibility of roof top rainwater harvesting potential—A case study of south indian university. *Cleaner Engineering and Technology*, 4, 100206.
- Baba, A., Tsatsanifos, C., El Gohary, F., Palerm, J., Khan, S., Mahmoudian, S. A., Ahmed, A. T., Tayfur, G., Dialynas, Y. G., & Angelakis, A. N. (2018). Developments in water dams and water harvesting systems throughout history in different civilizations. *International Journal of Hydrology*, 2(2), 150–166. <https://doi.org/10.15406/ijh.2018.02.00064>
- Bhattacharya, P. K. (2012). Sectional President's address: irrigation and agriculture in ancient India. *Proceedings of the Indian History Congress*, 73, 18–34. <https://www.jstor.org/stable/44156186> [Accessed March 10, 2023].
- Bid, S., Siddique, G., & Ghosh, S. (2018). Dam: Historical perspectives and an overview of India. *International Journal of Scientific Research and Review*, 7(1), 1–15.
- Bisht, R. S., Lal, B. B., & Gupta, S. P. (Eds.). (1984). *Structural Remains and Town Planning of Banawali*, in: *Frontiers of the Indus Civilization: Sir Mortimer Wheeler Commemoration Volume*. Books on behalf of Indian Archaeological Society jointly with Indian History Culture Society.
- Chakrabarti, D. K. (1995). *The Archaeology of Ancient Indian cities*. Oxford University Press.
- Chase, B., Ajithprasad, P., Rajesh, S., Patel, A., & Sharma, B. (2014). Materializing harappan identities: Unity and diversity in the borderlands of the indus civilization. *Journal of Anthropological Archaeology*, 35, 63–78.
- Dixit, Y., Hodell, D. A., Giesche, A., Tandon, S. K., Gázquez, F., Saini, H. S., Skinner, L. C., Mujtaba, S. A. I., Pawar, V., Singh, R. N., & Petrie, C. A. (2018). Intensified summer monsoon and the urbanization of Indus civilization in northwest India. *Scientific Reports*, 8(1), 1–12. <https://doi.org/10.1038/s41598-018-22504-5>
- Fardin, H. F., Hollé, A., Gautier, E., & Haury, J. (2013). Wastewater management techniques from ancient civilizations to modern ages: Examples from south asia. *Water Supply*, 13(3), 719–726.
- Feo, G. D., & Napoli, R. (2007). Historical development of the augustan aqueduct in southern italy: Twenty centuries of works from serino to naples. *Water Supply*, 7(1), 131–138.

- Ghani, U. (2019). A picture of the great bath of moenjodaro. [https://commons.wikimedia.org/wiki/File:A\\_picture\\_of\\_The\\_Great\\_Bath\\_of\\_Moenjodaro\\_by\\_Usman\\_Ghani.jpg](https://commons.wikimedia.org/wiki/File:A_picture_of_The_Great_Bath_of_Moenjodaro_by_Usman_Ghani.jpg) [Accessed February 25, 2023].
- Gray, H. F. (1940). Sewerage in ancient and mediaeval times. *Sewage Works Journal*, 12, 939–946.
- India.com (2015). Water reservoir at dholavira site. [https://commons.wikimedia.org/wiki/File:Water\\_reservoir\\_at\\_Dholavira\\_site.jpg](https://commons.wikimedia.org/wiki/File:Water_reservoir_at_Dholavira_site.jpg) [Accessed February 25, 2023].
- Iyer, M. (2019). The best laid plans. *Deccan Herald*.
- Jain, S. K., Lohani, A. K., Khobragade, S. D., Singh, P. K., & Ullah, M. F. (2018). *Hydrological Knowledge in Ancient India*. National Institute of Hydrology, Ministry of Water Resources, River Development and Ganga Rejuvenation, Roorkee, Uttarakhand, India.
- Jansen, M. (1989). Water supply and sewage disposal at mohenjo-daro. *World Archaeology*, 21(2), 177–192.
- Jha, P. A. (Ed.). (1988). *Vrhat Sanhita (550 AD) by Varahmihira*. Chow Khamba Vidyabhawan, Varanasi.
- Kenoyer, J. M. (1991). The indus valley tradition of pakistan and western india. *Journal of World Prehistory*, 5(4), 331–385.
- Kenoyer, J. M. (2003). Uncovering the keys to the lost Indus cities. *Scientific American*, 289(1), 66–75. <https://www.jstor.org/stable/26060364> [Accessed March 10, 2023].
- Kumar, A. (2014). *Grand Anicut, Kallanai* [Photograph]. Flickr. <https://www.flickr.com/photos/ashwinkumar/11875625145>
- Kumar, S., & Katara, S. (2020). Ancient knowledge of water and water traditions in india. *The Discussant*, VIII(4), 38–48.
- Lofrano, G., & Brown, J. (2010). Wastewater management through the ages: A history of mankind. *Science of The Total Environment*, 408(22), 5254–5264.
- Mane, S. P., & Shinde, A. S. (2014). A study of changing pattern of rainwater harvesting management from ancient to modern age in India: A geographical analysis. *Review of Research*, 3(10), 1–6.
- Mckean, M. B. (1985). *The Palynology of Balakot, a Pre-Harappan and Harappan Age Site in Las Bela, Pakistan*. PhD thesis, Southern Methodist University, Dallas, Texas, USA.
- Mun, J., & Han, M. (2012). Design and operational parameters of a rooftop rainwater harvesting system: Definition, sensitivity and verification. *Journal of Environmental Management*, 93(1), 147–153.
- Nair, K. S. (2004). Role of water in the development of Civilization in India—A review of ancient literature, traditional practices and beliefs. *International Association of Hydrological Sciences*, 286.
- Nigam, R., Dubey, R., Saraswat, R., Sundaresh, Gaur, A. S., & Loveson, V. J. (2016). Ancient Indians (Harappan settlement) were aware of tsunami/storm protection measures: A new interpretation of thick walls at Dholavira, Gujarat, India. *Current Science*, 111(12), 2040–2043. <https://www.jstor.org/stable/24911592> [Accessed March 10, 2023].
- Global Atlas of Environmental Justice. (2022). Rampur hydroelectric project. [https://ejatlas.org/media/conflict/rampurhydroelectric-project-india/rampur\\_contruction.png](https://ejatlas.org/media/conflict/rampurhydroelectric-project-india/rampur_contruction.png) [Accessed February 25, 2023].
- Ortloff, C. R. (2009). *Water engineering in the ancient world: Archaeological and climate perspectives on societies of ancient South America, the Middle East, and South-East Asia*. Oxford University Press.
- Probe, T. (2022). Subansiri lower hydroelectric project. [https://img-cdn.thepublive.com/filters:format\(webp\)/theprobe/media/post\\_attachments/wp-content/uploads/2022/06/166-thum-O.jpg](https://img-cdn.thepublive.com/filters:format(webp)/theprobe/media/post_attachments/wp-content/uploads/2022/06/166-thum-O.jpg) [Accessed February 25, 2023].
- Pruthi, R. (2004). *Prehistory and Harappan civilization*. APH Publishing.
- Sarasvati, S. P. (2009). *Rig Veda, DAV publication division*. New Delhi.
- Scarborough, V. L. (2003). *The flow of power: Ancient water systems and landscapes* (1st ed.). School of American Research Press, Santa Fe, NM, 1 edition.
- Sharma, S., & Shruthi, M. S. (2017). Water in Hindu scriptures: Thank you, water! In K. V. (Ed.), *Water and scriptures: Ancient roots for sustainable development*.
- Shaw, J., & Sutcliffe, J. (2001). Ancient irrigation works in the Sanchi area: An archaeological and hydrological investigation. *South Asian Studies*, 17(1), 55–75.
- Singh, P. K., Dey, P., Jain, S. K., & Mujumdar, P. P. (2020). Hydrology and water resources management in ancient India. *Hydrology and Earth System Sciences*, 24(10), 4691–4707.
- Sorcinelli, P. (1998). *Storia Sociale dell'acqua: Citi e culture*. Pearson Italia SpA.

- Srinivasan, T. M. (1970). Water lifting devices in ancient India: Their origin and mechanisms. *Indian Journal of History of Science*, 5, 1–15.
- Srinivasan, T. M. (1976). Measurement of rainfall in ancient India. *Indian Journal of History of Science*, 11(2), 148–157.
- Webster, C. (1962). The sewers of mohenjo-daro. *Journal of Water Pollution Control*, 34(4), 426–432.
- Wikipedia.org. (2008). Tehri dam. [https://en.wikipedia.org/wiki/Tehri\\_Dam#/media/File:Tehri\\_dam\\_india.jpg](https://en.wikipedia.org/wiki/Tehri_Dam#/media/File:Tehri_dam_india.jpg) [Accessed February 25, 2023].
- Wikipedia.org. (2010). The drainage system at lothal. [https://en.wikipedia.org/wiki/Lothal#/media/File:The\\_drainage\\_system\\_at\\_Lothal\\_2.JPG](https://en.wikipedia.org/wiki/Lothal#/media/File:The_drainage_system_at_Lothal_2.JPG) [Accessed February 25, 2023].
- Wikipedia.org. (2018). Sardar sarovar dam. [https://en.wikipedia.org/wiki/Sardar\\_Sarovar\\_Dam#/media/File:Sardar\\_Sarovar\\_Dam\\_1.jpg](https://en.wikipedia.org/wiki/Sardar_Sarovar_Dam#/media/File:Sardar_Sarovar_Dam_1.jpg) [Accessed February 25, 2023].
- Wolfe, P. (1999). *History of wastewater: World of water 2000—The past, present and future*. Water World, Water Wastewater Int. Suppl.
- Wright, R. P. (2010). *The ancient Indus: Urbanism, economy and society in South Asia*. Cambridge University Press.
- Yashwant, S. (2019). India's water wisdom in times of climate crisis. <https://indiaclimatedialogue.net/2019/10/11/indias-water-wisdom-in-times-of-climate-crisis/> [Accessed February 25, 2023].